

Connecting Numerical, Graphical and Algebraic Ideas of a Limit

Limits from tables of values, from graphs, from algebra, and as the sum of a series

Directions:

- Give exact values (fractions, not decimals) for every limit and every sum.
- Where a problem asks for a table value, round to four decimal places.
- The four approaches are mixed on purpose after problems 1–2: decide which view the problem is asking for before you start.

1. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ algebraically.

Answer: _____

2. Let $f(x) = \frac{x - 4}{\sqrt{x} - 2}$.

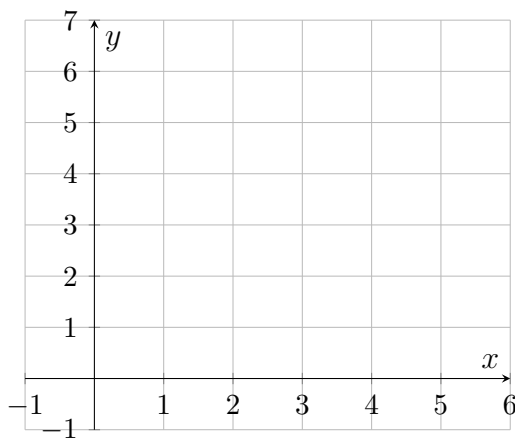
- (a) Compute $f(3.9)$ to four decimal places. _____
- (b) Compute $f(4.1)$ to four decimal places. _____
- (c) Give the exact value of $\lim_{x \rightarrow 4} f(x)$. _____

3. Let $g(x) = \begin{cases} x + 1, & x < 2 \\ 6 - x, & x \geq 2 \end{cases}$

(a) Find $\lim_{x \rightarrow 2^-} g(x)$. _____

(b) Find $\lim_{x \rightarrow 2^+} g(x)$. _____

(c) Sketch g on the grid, marking open and closed endpoints, and say whether $\lim_{x \rightarrow 2} g(x)$ exists.



4. Evaluate $\lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x + 1}$.

Answer: _____

5. Consider the geometric series $\sum_{n=1}^{\infty} 3 \left(\frac{1}{2}\right)^{n-1}$.

(a) Find the fourth partial sum S_4 as an exact fraction. _____

(b) Find the sum of the whole series. _____

6. Let $h(x) = \frac{\sin x}{x}$, with x in radians.

(a) Compute $h(0.1)$ to four decimal places. _____

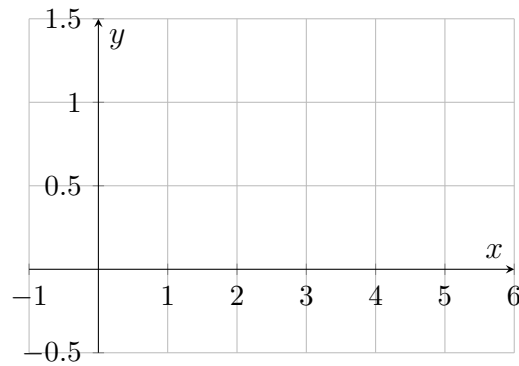
(b) Compute $h(0.5)$ to four decimal places. _____

(c) Give the exact value of $\lim_{x \rightarrow 0} h(x)$. _____

7. Let $p(x) = \frac{x-2}{x^2-4}$.

(a) Find $\lim_{x \rightarrow 2} p(x)$ as an exact fraction. _____

(b) Sketch p near $x = 2$ on the grid, marking what happens at $x = 2$ itself, and say whether the graph has a hole or a vertical asymptote there.



8. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$ as an exact fraction. (Multiplying by the conjugate helps.)

Answer: _____

9. Consider $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$.

- (a) Find the sum of the series. _____
- (b) Write the partial sum S_N in closed form and explain why the sum of the series is a limit rather than an addition.

10. Let $q(n) = \left(1 + \frac{1}{n} \right)^n$.

- (a) Compute $q(100)$ to four decimal places. _____
- (b) Compute $q(10000)$ to four decimal places. _____
- (c) Name the number these values are approaching, and explain why a table alone cannot prove that is the limit.

11. Consider $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$.

- (a) Find the limit. _____
- (b) Explain why direct substitution does not settle this, and why cancelling the common factor is legitimate inside a limit.

12. Synthesis. Let $r(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$

- (a) Find $\lim_{x \rightarrow 1} r(x)$. _____
- (b) Sketch r on the grid and explain why r is not continuous at $x = 1$ even though the limit there exists.

